Research proposal: ‘Towards Perfect Code Generation: Iterative API-based Feedback Loops’

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Abstract

Background

Large Language Models (LLMs) are very capable in generating code when prompted to do so. Their ability to generate code that looks correct and compiles as expected does not necessarily mean it always follows best practices. LLMs are trained on real code snippets, which might not be entirely semantically correct. A bad habit of novice programmers when writing HTML code is to overuse the `<div>` tag to define containers, instead of other, more semantically meaningful tags like `<section>`, `<article>`, or `<nav>`. An overuse of the `<div>` tag makes the overall codebase less maintainable and also negatively impacts the accessibility of a website for users using screen readers. An example of this can be found in Appendix A (Listing 1).

Research shows that only a small number of samples can poison LLMs of any size (Souly et al., 2025), which means that this bad habit, nicknamed ‘divitis’, is very likely to have poisoned the training data of LLMs. Code which was affected by ‘divitis’ looks correct, does not result in any compilation errors, and a developer that does not know about the accessibility issues might not even notice the problem. You can configurate an LLM to take these best practices into account when generating code, but it is not guaranteed that it will always follow its system prompt’s instructions, especially when running local or smaller models. There exist several Application Programming Interfaces (APIs) online that can be used to check the quality of code. One example is the W3C Markup Validation Service[1].

We propose to use such APIs to create a guardrail for LLM outputs which consists of an iterative feedback loop that will continously find errors in the code and regenerate the code by passing the errors the API found in a ‘regeneration prompt’.

Research questions

Some text. The study will address the following research questions:

[1] `https://validator.w3.org/`

RQ1. First question

RQ2. Second question

Write some more text before heading into the research question below

Our research question is as follows: *"To what extent is it possible to use this guardrail to generate semantically correct HTML code that also adheres to best practices?"*.

Research methods

Appendix A
Code Listings

```html
<!-- Generic header -->
<div class="header">
  <div class="logo">My Dev Blog</div>
  <!-- This should be a <nav> -->
  <div class="menu">
    <div class="link">Home</div>
    <div class="link">About</div>
  </div>
</div>

<!-- Main content area -->
<div class="main-content">

  <!-- This should be an <article> -->
  <div class="post">
    <div class="title">How to Write Clean Code</div>
    <div class="content">Using semantic tags is a great start...</div>
  </div>

  <!-- This should be a <section> -->
  <div class="comments-area">
    <div class="subtitle">Comments</div>
    <div class="comment">Great post!</div>
  </div>

</div>
```

Listing 1: Example of ‘divitis’ (an overuse of the `<div>` tag) in HTML code.

Appendix B
API Error Messages

something something something

References

Souly, A., Rando, J., Chapman, E., Davies, X., Hasircioglu, B., Shereen, E., ... Kirk, R. (2025). *Poisoning attacks on llms require a near-constant number of poison samples.* Retrieved from https://arxiv.org/abs/2510.07192